

WASP-12b

R-0083

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_251227 · created 2026-07-15T21:21:41+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461036.7543 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.002 +/- 0.045
Transit depth (Rp/Rs)^2	0.00055 +/- 0.021 [%]
Orbital Inclination (inc)	83.78 +/- 2.91
Airmass coefficient 1 (a1)	0.76 +/- 0.42
Airmass coefficient 2 (a2)	0.47 +/- 1.1
Scatter in the residuals of the lightcurve fit is	52.98 %
Best Comparison Star	#3 - [255, 198]
Optimal Aperture	3.41
Optimal Annulus	5.12
Transit Duration (day)	0.1099 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.002 ± 0.045 vs 0.1178 (deviation 2.55σ)
Duration fit vs published	0.1099 d vs 0.125 d (deviation 3.05σ)
Mid-time O–C	-2.6 min
Depth SNR	0.0
Residual scatter	52.98 %

Light curve

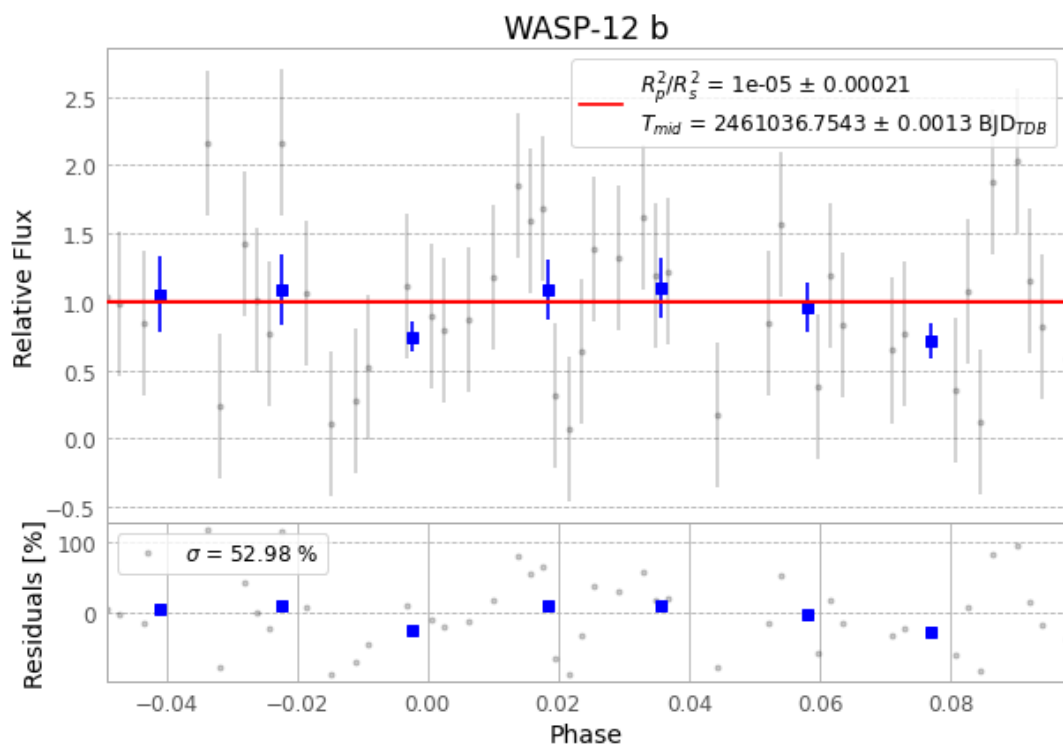


Figure 1 — FinalLightCurve_WASP-12 b_2025-12-26.png (SHA-256 5da7c1b984e5764a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [628, 408], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bc858debb5e7c70d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

82 FITS frames (54 MB), WASP-12251227043015.FITS → WASP-12251227083316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-251227.log (a1bd2a267941...), FinalLightCurve_WASP-12 b_2025-12-26.png (5da7c1b984e5...), FinalLightCurve_WASP-12 b_2025-12-26.csv (2e0c26d800e1...)

Compute resources

Wall time 270.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 21:21Z.